

Necessity of Right-Handed Weyl Neutrinos for Lepton Mass Generation in Vector-Like Regularized Standard Models: A Theoretical Validation

Input Question

What are the smallest building blocks of matter?

Domain

Physics

Validation Status

needs_data

Problem Statement

The user asks for the fundamental, indivisible constituents of matter. However, the allowed evidence sources (EV-Q075-29b40f883323d60fe3abcc0d and EV-Q075-5c8e119ab0a790ec4aad171b) are specialized research papers focusing on flavor symmetries and lepton mass generation mechanisms, respectively. They do not contain general definitions of the Standard Model's fundamental particles (quarks and leptons) as the 'smallest building blocks'. Consequently, there is insufficient evidence in the allowed set to directly answer the general question. This plan addresses a valid, narrow theoretical sub-problem derived from the available evidence: the necessity of right-handed Weyl neutrinos for lepton mass generation in vector-like regularized Standard Models.

Rationale

Directly answering 'quarks and leptons' would require citing general textbooks or review articles which are not in the allowed evidence list. The booklet excerpt contains this information but is explicitly prohibited as an evidence source. Therefore, we must pivot to a hypothesis strictly supported by the allowed arXiv papers.

EV-Q075-5c8e119ab0a790ec4aad171b provides a specific theoretical framework where right-handed neutrinos are necessary for mass generation. Investigating this mechanism allows us to produce a scientifically valid output grounded in allowed evidence, while explicitly acknowledging the knowledge gap regarding the general definition of matter's building blocks.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: In the vector-like regularized Standard Model framework described by EV-Q075-5c8e119ab0a790ec4aad171b, right-handed Weyl neutrinos are strictly necessary for consistent lepton mass generation, distinguishing this specific theoretical extension from the minimal chiral Standard Model.

- **Mechanism**: The vector-like regularization scheme modifies the ultraviolet behavior of the theory such that standard chiral symmetry breaking mechanisms are insufficient; the introduction of right-handed Weyl fields restores consistency in the lepton sector mass terms as derived in the cited work.
- **Falsifiable Prediction**: If a rigorous re-derivation using the specific vector-like regularization parameters from EV-Q075-5c8e119ab0a790ec4aad171b demonstrates that lepton masses can be generated without right-handed Weyl neutrinos, or if global fits to neutrino oscillation data (NuFit) definitively exclude the specific mass-mixing correlations predicted by this vector-like model at $>5\sigma$ confidence, the hypothesis is falsified.
- **Required Observations**: Analytical verification of the necessity proof in EV-Q075-5c8e119ab0a790ec4aad171b ; Mapping of vector-like regularization parameters to effective field theory operators ; Statistical comparison of vector-like model predictions against NuFit global fit data and rare decay limits
- **Risk of Being Wrong**: High. The hypothesis addresses only a niche theoretical sub-problem (lepton mass in regularized SM) and does not answer the user's core question about general 'smallest building blocks'. Furthermore, the mapping between the 1997 regularization formalism and modern precision observables is a significant knowledge_gap that may prove theoretically intractable.

Hypothesis 2

- **Hypothesis**: Rare decay flavor symmetry patterns in quark and lepton sectors, as analyzed in EV-Q075-29b40f883323d60fe3abcc0d, provide sufficient constraints to distinguish between fundamental point-like particles and composite substructures at currently accessible energy scales.
- **Mechanism**: Flavor symmetries act as selection rules; deviations in rare decay rates or angular distributions from Standard Model predictions would signal either new physics or compositeness, thereby indirectly probing the 'building block' nature of fermions through their interaction signatures rather than direct imaging.
- **Falsifiable Prediction**: If rare decay measurements across multiple channels (e.g., B-meson, Kaon decays) remain consistent with Standard Model predictions within experimental uncertainties, and no flavor symmetry violations attributable to substructure are observed, the hypothesis that these specific observables can probe fundamentality is weakened.
- **Required Observations**: Precision measurements of rare quark and lepton decay branching ratios ; Analysis of flavor symmetry violation patterns consistent with EV-Q075-29b40f883323d60fe3abcc0d methodology ; Exclusion of alternative new physics explanations for any observed anomalies
- **Risk of Being Wrong**: Very High. The evidence EV-Q075-29b40f883323d60fe3abcc0d focuses on flavor symmetries and rare decays but does not explicitly link these to testing 'fundamentality' vs 'compositeness'. This hypothesis extrapolates beyond the provided evidence scope.

Technical Details

This research plan addresses a narrow theoretical sub-problem derived from the allowed evidence, explicitly acknowledging that the original user question regarding the 'smallest building blocks of matter' cannot be answered due to insufficient_evidence in the provided sources. The plan focuses strictly on validating the hypothesis that 'In the vector-like regularized Standard Model described by EV-Q075-5c8e119ab0a790ec4aad171b, right-handed Weyl neutrinos are required for consistent lepton mass generation.' The technical approach involves: (1) Analytical re-derivation of the mass generation mechanism using the specific vector-like regularization scheme cited in EV-Q075-5c8e119ab0a790ec4aad171b to confirm the

necessity of right-handed Weyl fields; (2) Mapping the resulting theoretical parameters to effective field theory (EFT) operators to identify observable consequences; (3) A high-risk knowledge_gap identification step to establish whether these theoretical parameters can be constrained by external datasets (NuFit/Rare Decays), noting that this mapping is not supported by the allowed evidence and requires independent theoretical justification.

Datasets

Source

```
```json
[
{
 "id": "EV-Q075-5c8e119ab0a790ec4aad171b",
 "type": "theoretical_framework",
 "description": "ArXiv paper (hep-ph/9705325) discussing the necessity of right-handed Weyl neutrinos due to vector-like phenomenon in regularized SM."
}
]
```
```

Target

```
```json
[
{
 "name": "NuFit_Global_Fit_Data",
 "type": "external_experimental_constraints",
 "description": "Global fit data for neutrino oscillation parameters. Note: Usage assumes a validated theoretical mapping from vector-like regularization parameters to oscillation observables, which is currently a knowledge_gap.",
 "version": "latest_available",
 "source_url": null
},
{
 "name": "Rare_Decay_Limits",
 "type": "external_experimental_constraints",

```

```
"description": "Upper limits on rare lepton-flavor-violating decays. Note: Usage assumes a validated theoretical mapping, which is currently a knowledge_gap.",
"version": "latest_available",
"source_url": null
}
]
'''
```

## Paper Abstract

Background: The question of matter's fundamental building blocks is typically answered by the Standard Model's quarks and leptons. However, allowed evidence sources lack general definitions, necessitating a focus on specific theoretical mechanisms. Methods: We investigate the vector-like regularized Standard Model framework presented in EV-Q075-5c8e119ab0a790ec4aad171b, specifically examining the role of right-handed Weyl neutrinos in lepton mass generation. We perform an analytical re-derivation of the mass term constraints to verify the logical necessity of these fields within the specified regularization scheme. Validation Plan: The derivation will be cross-checked using symbolic computation tools. We explicitly identify the mapping between regularization parameters and observable phenomena (e.g., NuFit data) as a critical knowledge gap. Results: pending (待执行验证实验).

## Methods

1. Theoretical Reconstruction: Perform symbolic algebra derivation to reproduce the mass term constraints from EV-Q075-5c8e119ab0a790ec4aad171b, verifying the logical necessity of right-handed Weyl neutrinos within that specific regularization framework. 2. Knowledge Gap Analysis: Explicitly document the lack of evidence linking the vector-like regularization parameters to standard EFT operators used in NuFit/Rare Decay analyses. This step identifies the 'mapping' as a critical unverified assumption. 3. Phenomenological Feasibility Study (Pending): Outline the Bayesian inference framework that would be used IF the mapping were established, comparing the vector-like model against the Minimal Standard Model. This step remains theoretical due to the missing mapping evidence.

## Experiments

### Baselines

```
'''json
[
 "Minimal Standard Model (MSM) with massless neutrinos",
 "Standard Model with ad-hoc Majorana mass terms (See-Saw Mechanism Type I)"
]
'''
```

## Metrics

```
```json
```

```
[
```

```
"Logical Consistency Check (Pass/Fail) of the re-derived mass term against  
EV-Q075-5c8e119ab0a790ec4aad171b",
```

```
"Theoretical Mapping Completeness Score (Qualitative assessment of the gap between regularization params and  
observables)",
```

```
"Bayesian Evidence Ratio (Placeholder for future execution if mapping is resolved)"
```

```
]
```

```
```
```

## Ablation

Remove the vector-like regularization constraint from the derivation to test if the necessity of right-handed neutrinos persists in standard chiral regularization schemes, thereby isolating the specific contribution of the cited framework.

## Validation Protocol

Cross-check the analytical derivation with independent symbolic computation tools (e.g., SymPy vs Mathematica). Validate that the hypothesis scope is strictly limited to 'lepton mass generation in regularized SM' and does not overclaim fundamentality.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- **\*\*EV-Q075-5c8e119ab0a790ec4aad171b\*\*** · arxiv · arXiv:hep-ph/9705325
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/hep-ph/9705325.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED; locator=page:1|section:page-1|paragraph:1; content\_sha256=136034761e52c5cd5babe265f8efa85d3dfec165bb0626387dc98fe0176fc0b7

## Reviewer Comments

- The revised hypothesis correctly restricts its scope to 'lepton mass generation in the vector-like regularized SM' as supported by EV-Q075-5c8e119ab0a790ec4aad171b, successfully resolving the previous overgeneralization regarding 'fundamental structure of matter'.

- The ResearchPlan now explicitly declares 'insufficient\_evidence' for the original user question ('smallest building blocks') and frames the current work as a valid, narrow theoretical sub-problem, adhering to system constraints.
- The experiment design appropriately identifies the mapping between vector-like regularization parameters and NuFit/Rare Decay observables as a 'high-risk knowledge\_gap' rather than assuming it as established fact, which satisfies the requirement for epistemic honesty.
- Results are correctly marked as pending/non-executed. No fabricated data or unverified causal claims are present. References are strictly limited to allowed evidence IDs.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Verify analytical derivation matches EV-Q075-5c8e119ab0a790ec4aad171b exactly
- Explicitly document the 'knowledge\_gap' regarding the mapping between vector-like parameters and NuFit/Rare Decay observables
- Ensure hypothesis text strictly avoids claims about 'fundamental structure of matter' or 'smallest building blocks'
- Provide code for symbolic derivation with fixed random seeds where applicable
- Document all assumptions regarding heavy neutrino mass scales as unverified